

Bayesian Neural Networks for Multiclass Classification

General Principles

Building upon the binary classification [Bayesian Neural Network \(BNN\)](#), multiclass classification extends the logic to scenarios where an observation belongs to one of K mutually exclusive categories ($K > 2$).

Instead of the final layer returning a single output, the final layer in a multiclass BNN returns a K -dimensional vector of scores (logits) for each observation. To transform these continuous scores into valid probabilities that sum to 1 across all K classes, we apply the **softmax** activation function. Finally, the categorized predictions are evaluated using a **Categorical** likelihood.

Considerations

Note

- **Output Layer Dimensions:** While binary classification network predictions can be compressed to a single output logit per observation, multiclass networks **MUST** output exactly K dimensions in their final layer, matching the number of target classes.
- **The Softmax Simplex:** Applying the softmax function across the final layer's logits guarantees that the resulting outputs form a probability simplex . This is biologically similar to independent Poisson rates strictly normalizing to fixed categorical ratios.
- **Likelihood Function:** After calculating the probabilities with softmax, we use a **Categorical** distribution as the final likelihood, matching the integer index of the observed category.
- **Improved Calibration:** Multiclass BNNs greatly reduce out-of-distribution over-confidence. Standard deep learning cross-entropy models will often assign $>99\%$

probability to an unseen class purely due to the exponential nature of softmax. In a BNN, exploring the posterior width of the parameters yields “flat” unconfident probability profiles over K classes when the input is outside the training distribution.

Example

Below is an example code snippet demonstrating a *Bayesian Neural Network for multiclass classification* using the Bayesian Inference (BI) package. This example generates a synthetic $K = 3$ cluster dataset.

Python

```
from BI import bi
import jax.numpy as jnp
import jax

# Setup device-----
m = bi(platform='cpu')

# Generate Synthetic Data -----
# 3 classes based on a random normal distribution split
key = jax.random.PRNGKey(42)
X = jax.random.normal(key, (300, 2))
# Rule: Q1=Class 0, Q2/Q3=Class 1, Q4=Class 2
Y = jnp.where(X[:, 0] > 0, jnp.where(X[:, 1] > 0, 0, 1), 2)

m.data_on_model = dict(X=X, Y=Y)

# Define model -----
def model(X, Y, D_H1=5, K=3):
    N, D_X = X.shape

    # First hidden layer: 2 input features -> 5 hidden units
    w1 = m.bnn.layer_linear(
        X,
        dist=m.dist.normal(0, 1, name='w1_weight', shape=(D_X, D_H1)),
        activation='tanh'
    )
```

```

# Final output layer: 5 hidden units -> K output units
# Note: No activation is applied automatically inside the layer function here
w2 = m.bnn.layer_linear(
    w1,
    dist=m.dist.normal(0, 1, name='w2_weight', shape=(D_H1, K))
)

# Apply Softmax across the K dimension (axis=-1) to yield probabilities
p = jax.nn.softmax(w2, axis=-1)

# Categorical Likelihood matching indices in Y
m.dist.categorical(probs=p, obs=Y)

# Run mcmc -----
m.fit(model) # Approximate posterior distributions

# Predictions from the model -----
import matplotlib.pyplot as plt

# Create a grid to evaluate the model
n_grid = 50
x0 = jnp.linspace(X[:, 0].min() - 0.5, X[:, 0].max() + 0.5, n_grid)
x1 = jnp.linspace(X[:, 1].min() - 0.5, X[:, 1].max() + 0.5, n_grid)
xx0, xx1 = jnp.meshgrid(x0, x1)
X_grid = jnp.c_[xx0.ravel(), xx1.ravel()]

# Swap data on model temporarily to predict on the grid
m.data_on_model = dict(X=X_grid, Y=jnp.zeros(X_grid.shape[0], dtype=jnp.int32))
pred = m.sample(data = m.data_on_model)['x']
p_mean = jnp.mean(pred, axis=0)

# Plotting the posterior predictive mean (categorical blending)
fig, ax = plt.subplots(figsize=(8, 6), constrained_layout=True)
contour = ax.contourf(xx0, xx1, p_mean.reshape(n_grid, n_grid), cmap="viridis", alpha=0.6)
scatter = ax.scatter(X[:, 0], X[:, 1], c=Y, cmap="viridis", edgecolors='k')
ax.set(title="Posterior Predictive Mean", xlabel="Feature 1", ylabel="Feature 2")
fig.colorbar(contour, ax=ax)

```

jax.local_device_count 16

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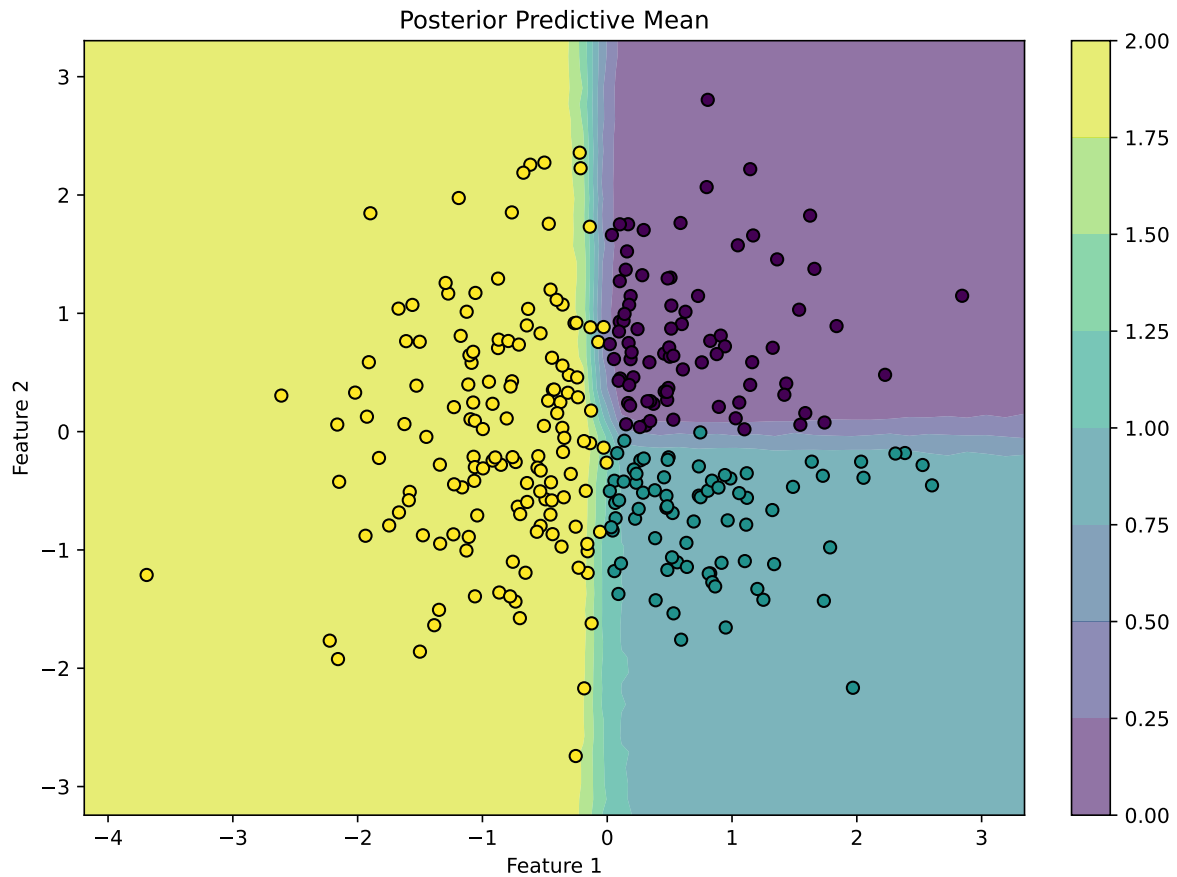
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warmup: 0%| | 1/1000 [00:01<31:22, 1.88s/it, 1 steps of size 2.34e+00. acc. prob.
C:\Users\Sosa\Documents\BI\.venv\Lib\site-packages\BI\Main\main.py:439: UserWarning:

Sample's batch dimension size 500 is different from the provided 1 num_samples argument. Def

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Julia

```
using BayesianInference
using PythonCall

# Setup device-----
m = importBI(platform="cpu")

# Generate Synthetic Data -----
np = pyimport("numpy")
jax_random = pyimport("jax.random")
jnp = pyimport("jax.numpy")

key = jax_random.PRNGKey(42)
X = jax_random.normal(key, (300, 2))

# Simple rule to partition into K=3 classes
Y = jnp.where(X[:, 0] > 0, jnp.where(X[:, 1] > 0, 0, 1), 2)

m.data_on_model["X"] = X
m.data_on_model["Y"] = Y

# Define model -----
@BI function model(X, Y)
    N, D_X = size(X)
    D_H1 = 5
    K = 3

    # First hidden layer
    w1 = m.bnn.layer_linear(
        X,
        dist=m.dist.normal(0, 1, name="w1_weight", shape=(D_X, D_H1)),
        activation="tanh"
    )

    # Final output layer
    w2 = m.bnn.layer_linear(
        w1,
        dist=m.dist.normal(0, 1, name="w2_weight", shape=(D_H1, K))
    )

    # Softmax conversion to probability simplex
```

```

p = jax.nn.softmax(w2, axis=-1)

# Categorical Likelihood
m.dist.categorical(probs=p, obs=Y)
end

# Run mcmc -----
m.fit(model, num_samples=500, progress_bar=false)

```

Mathematical Details

Bayesian Formulation

For a multiclass classification task spanning N observations and K mutually exclusive classes, we model the probability vector θ_i that the response $Y_i \in \{0, 1, \dots, K - 1\}$ falls into each respective class.

1. Hidden Layer:

$$\Omega_{i,1} = \tanh(X_i \Theta_1)$$

Where X_i is a 2-vector of features and Θ_1 is a 2×5 weight matrix parameterized by a Standard Normal prior: $\Theta_1 \sim \text{Normal}(0, 1)$.

2. Output Layer (Logits):

$$\phi_i = \Omega_{i,1} \Theta_2$$

Where Θ_2 is a 5×3 weight matrix mapping the hidden features to the logits for the $K = 3$ classes. This is parameterized by: $\Theta_2 \sim \text{Normal}(0, 1)$.

3. Softmax Transformation:

$$\theta_i = \text{Softmax}(\phi_i)$$

The softmax function ensures that for every observation i , the sum of probabilities across the 3 classes equals exactly 1.

4. Likelihood:

$$Y_i \sim \text{Categorical}(\theta_i)$$

The observed class index Y_i is drawn from a Categorical distribution parameterized by the probability vector θ_i .

Notes

Note

- For large outputs where $K > 100$, computing the exact softmax normalization scalar (the denominator term combining all exponentiated logits) can become computationally expensive over thousands of MCMC posterior evaluations.
- Neural networks configured with a standard Cross-Entropy loss mapping to one-hot vectors conceptually perform exactly this sequence: dot product of final weights $\rightarrow$ Softmax $\rightarrow$ Categorical Likelihood.

Reference(s)

1. [PyData Berlin 2025: Introduction to Stochastic Variational Inference with NumPyro](#)